Climatic Area	Standard Mixes (by weight)	Application
Northern **(Terrace Area)**	**North West General Mix** Creeping red fescue 50% Timothy 8% Kentucky bluegrass 7% Hard fescue/Sheep fescue 35%	General seeding inland where mean annual precipitation is > 50 cm. For use in CWH and ICH biogeoclimatic zone. Fertilizer: 22-11-11, or pre-approved equivalent
	North West Dryland Mix Intermediate wheatgrass 61% Crested wheatgrass 23% Hard fescue/Sheep fescue 15% Kentucky bluegrass 1%	General seeding inland where mean annual precipitation is < 50 cm. For use in SBS biogeoclimatic zone (East of Moricetown to West of Endako) Fertilizer: 22-11-11, or pre-approved equivalent
	Northern Coastal Mix Intermediate wheatgrass 52% Kentucky bluegrass 12% Hard fescue/Sheep fescue 27% Timothy 9%	General seeding coastal locations where mean annual precipitation is > 90 cm. For use in CWH biogeoclimatic zone (Haida Gwii, Prince Rupert to Pacific) Fertilizer: 22-11-11, or pre-approved equivalent
	Northern Mix Crested wheatgrass 35% Creeping red fescue 35% Kentucky bluegrass 30%	General seeding coastal locations where mean annual precipitation is > 90 cm. For use in ICH and BWBS biogeoclimatic zones: • (ICH – Pacific to Moricetown, Kitwanga to Thomas Creek 220 km N) • (BWBS – Thomas Creek to Yukon border) Fertilizer: 22-11-11, or pre-approved equivalent
	Ditch Vegetation Seed Mixture Crested wheatgrass 67% Creeping meadow foxtail 8% Kentucky bluegrass 3% Tufted hairgrass 22%	For use in revegetating roadside ditches following ditch maintenance operations. Fertilizer: 22-11-11, or pre-approved equivalent
Vancouver Island	**Vancouver Island / Coast Mix** Canada bluegrass 5% Creeping red fescue 29% Hard fescue/Sheep fescue 17% Perennial ryegrass 37% Redtop 3% Timothy 9%	General seeding coastal locations where mean annual precipitation is > 90 cm. Fertilizer: 18-18-18
Riparian Areas	**Riparian Area Mix** Slender wheatgrass 40% Perennial rye 25% Kentucky bluegrass 15% Timothy 10% Redtop 5% Junegrass 5%	General purpose for riparian or saturated soil areas throughout the Province. Fertilizer: 18-18-18

SECTION 766
IRRIGATION

DESCRIPTION

766.01 Scope – The work consists of the supply of all material, labour and equipment to install a complete and operating irrigation system as shown on the Drawings. The Contractor shall be responsible for obtaining all permits required.

766.02 References – Codes & Standards – In the absence of other instructions, the provisions of all the following codes and standards shall apply:

- The National Building Code of Canada;
- Current CSA Specifications for copper, steel and plastic pipe;
- AWWA Specification, current editions.

766.03 Site Security – The Contractor shall be responsible for maintaining all security at the project Site at all times, and shall ensure that no loss, damage or breakage occurs to stockpiled materials or to the partially installed irrigation system. The Contractor shall make good all damage resulting from acts of vandalism throughout the period of installation and subsequent maintenance.

766.04 Electrical Trades – The Contractor shall contact the Electrical Trades Supervisor prior to performing work on any existing Ministry owned equipment. All AC electrical connections shall be done by a qualified electrician.

766.05 Drawings and Instructions – The Contractor shall install the irrigation system as shown and detailed on the Drawings and Specifications. No deviation from the indicated make and model or installed location of valves, mains, laterals or any other irrigation system component will be allowed without first obtaining written permission from the Ministry Representative.

The Contractor shall maintain a daily record of construction activities. Upon completion of the Contract, the Contractor shall incorporate all accumulated information relevant to the Contract into the required as-built drawings. The as-built drawings shall be to the Ministry Representative's satisfaction, shall be reproducible, and shall be compiled by a competent professional draftsperson.

The Contractor shall prepare drawings showing the final location and make of all heads, emitter locations, pipe layout and other pertinent information, and shall submit two sets of these drawings to the Ministry Representative. The Contractor shall prepare another Drawing showing the wiring and automatic controller station numbers with all electrical data. The Contractor shall submit two copies of this Drawing and one set of the operating instructions for the controller, complete with spare parts list to the Ministry Representative. All drawings, spare parts lists, and operating instructions shall be Cerlox bound into a plastic covered 8½" x 11" booklet.

After the system has been completed, the Contractor shall instruct the Ministry Representative's agent in the proper use of the equipment.

Completion will not be certified until adjustments and drawings are approved.

MATERIALS

766.11 General – Shipping, handling and installation of materials shall be to manufacturer's recommended instructions, and best work practice. Particular care shall be taken to avoid scratches and nicks on the plastic pipe. Pipe must be properly stacked and stored in a clean place on the Site, keeping dirt out of the pipe at all times.

766.12 Pipes and Fittings – Galvanized pipe, Schedule 40, with galvanized fittings, shall be used inside culverts. The pipe shall be connected to the plastic main or lateral 450 mm clear of the culvert.

Plastic pipe shall be used for the submain and laterals of the irrigation system. Plastic pipe shall be semi-rigid extruded from PVC (Polyvinyl Chloride) resin, Type 1, grade 2, normal impact.

The minimum classes to be used are listed in Table 766-A.

Fittings shall be PVC plastic, Schedule 40 or 80, designed for solvent welding to PVC pipe. All fittings must have 1/2 to 2/3 interface fit to ensure a fully seated joint. Individual fittings shall be selected to ensure a proper fit, or they will be rejected.

All pipe and fittings shall be continuously and legibly marked with at least the following information:

- Manufacturer's name or trademark;
- Pressure rating;
- Type of material.

Pipe that is not marked to the satisfaction of the Ministry Representative will be rejected and shall be removed from the Site by the Contractor.

Table 766-A: Minimum Classes of Plastic Pipe

Up to 25 mm diameter	Class 200 for excavated trenches
Greater than 25 mm diameter	Class 160 for excavated trenches
All sizes	Class 200 for pipe flow

766.13 Cement – Pipe cement for solvent welding shall be of the type and make recommended by the pipe manufacturer, supplied to the Site in sealed containers clearly marked with the name of the manufacturer and the lot number. The Contractor shall comply with the manufacturer's instructions and safety procedures.

766.14 Sprinkler Heads – Risers for turf heads, lawn heads, impact heads, shrub heads and quick-coupling (QC) valves shall be swing joint type, fabricated from Schedule 80 PVC or Schedule 40 galvanised pipe and fittings, as detailed in the Specifications. Teflon tape shall be used on all threaded connections.

Sprinkler heads shall be as detailed on the Drawings.

766.15 Valves – Automatic valves shall be as detailed on the Drawings and the same make shall be used throughout. Automatic valves shall have flow control stems.

Valves shall be installed in Carson Industries No. 1419 valve boxes or other approved alternative complete with extensions and covers, as required. The top of all valve box covers shall be flush with the finished grade.

766.16 Controllers – Automatic controllers shall be supplied to operate the electrically controlled automatic valves. Controllers shall be 24-volt A.C. outlet, Class 2 rating, compatible with the valves used. Controllers and automatic valves shall be by the same manufacturer, unless otherwise noted. Controllers and transformers must bear CSA or Provincial stamps of approval. Controllers shall be as detailed on the Drawings.

Controllers shall be installed in Ministry standard controller box 30-A-120/240V, weatherproof, stainless steel service panel SN1765A as shown on SS Drawing SP635-2.4.8, or pre-approved equal complete with a Masterlock No. 15 padlock and two sets of keys for the lock.

766.17 Wiring – Wire between controllers and automatic valves shall be of a type approved for direct burial. Where control wires are exposed or pass through culverts, they shall be installed in rigid electrical conduit.

Wire shall be minimum 14-gauge single strand T.W.U.

766.18 Selected Native Fill – Native fill selected for backfilling shall be free draining, fine grained sand and/or gravel, containing no material larger than 19 mm in diameter, wood, or other debris; or may other approved fill material. All backfill shall be subject to the approval of the Ministry Representative.

CONSTRUCTION

766.31 General – Damaged Material – Damaged material shall be rejected on the decision of the Ministry Representative.

Plastic pipe, fittings, valves and appurtenances shall not be repaired by patching or taping. Where materials have been damaged, the damaged materials shall be removed and new parts or sections shall be installed.

766.32 Line Location – The Contractor shall ensure that all irrigation pre-ducts for passage of irrigation lines under roadways, medians, traffic islands and other surface impediments have been installed and are clearly marked at all entry points.

No irrigation line shall be installed parallel to and directly over another irrigation line or line of another trade. Lines laid in the same trench shall be a minimum of 50 mm apart. No pipe shall be installed closer than 300 mm to any parallel electric conduit as shown on SS Drawing SP635-1.5.2.

766.33 Compaction – Before laying the pipe, the Contractor shall be satisfied as to the extent of compaction in the lawn and planting areas.

766.34 Excavation – Excavated soil shall be carefully placed adjacent to the trench for convenient backfilling. Topsoil and subsoil shall be piled separately to avoid contamination of the topsoil.

Stones or other objects larger than 75 mm at their widest point shall be removed from the trenches. Holes below grade lines, caused by the removal of stones, must be filled in and compacted uniformly with the adjacent trench.

766.35 Laying the Pipe – Pipe shall be laid by trench excavation or by an approved vibrating pipe plough. Plastic pipe shall be laid on sand or selected native fill to a compacted depth of 50 mm. A further 75 mm of sand or selected native fill shall be placed over plastic pipes prior to backfilling. Pipes shall be run in straight lines between fittings. Pipe must not be supported at intermediate points on stones, bricks or other hard material.

All mains and laterals shall have a minimum cover of 400 mm of soil as measured from the top of the pipe to the finished grade.

Lawn and planting areas shall be disturbed as little as possible.

The Contractor shall manicure the finished grade over all mains and laterals upon completion of the pipe installation. All debris, rocks over 50 mm diameter, etc., that have been brought to the grade surface shall be removed to the Contractor's disposal location. The area over all trenches shall be fine graded and shall conform to SS 751.34.

766.36 Connections – The Contractor shall make connection to the existing water supply where shown on the Drawings. The Contractor shall ensure that the recommended operating pressure of the irrigation system is not exceeded by the water pressure at the source, by installing pressure regulators as required.

766.37 Inspection and Testing – After the pipe is in place in the bottom of the trench with risers in place, the risers shall be capped where the sprinklers will be attached, and all pipe fittings exposed. The maximum pressure shall be

applied to the system and maintained for a minimum of one hour.

All fittings shall be visually inspected and any that leak shall be cut out and replaced. Leaks shall not be repaired by patching or taping. The test pressure shall be maintained for one hour after replacing any defective sections. The section shall be re-inspected as before.

The system shall be flushed out to remove dirt and then the sprinklers shall be attached using Teflon tape or pre-approved non-setting pipe thread compound.

766.38 Backfill – After approval by the Ministry Representative, the trenches shall be backfilled, maintaining pressure in the line. If there is any indication of a leak, the defective section shall be located and replaced.

The trenches shall be carefully backfilled with the subsoil, followed by the topsoil. Both shall be compacted to the same density as the soil in the trench walls to minimize differential settlement. Backfill around turf heads with 0.03 m^3 of sand.

766.39 Controller Installation – The location of the controllers shall be determined on Site in the areas indicated on the Drawings. The Contractor shall have a qualified electrician connect the controllers to the electrical supply.

766.40 Adjustments – The sprinkler system shall be adjusted section by section to give satisfactory coverage to all areas. Pressure at the heads and/or Q.C. valves shall be as noted on the Drawings. Turf heads, lawn heads and Q.C. valves shall be set flush with the final turf grade by adjusting the swing joint riser, as required. During the landscape maintenance/guarantee period, the Contractor will return twice and adjust the heads, as required, to be flush with the final turf grade.

These call-backs shall be done within five days of notification by the Ministry Representative and shall be considered part of the Contract requirements.

766.41 Surplus Material – Surplus materials shall be removed from the Site.

766.42 Conditions for Acceptance – Completion will not be certified until adjustments are completed and as-built drawings prepared, approved, and bound into an approved booklet to the Ministry Representative's satisfaction. A copy of the inspection certificate issued by Technical Safety BC indicating compliance with the Electrical Code and a copy of the "Irrigation Systems Pressure Loss Calculation Data Sheet" found in APPENDIX 766-A (or alternative acceptable to the Ministry Representative) shall be included in the hardcover booklet.

766.43 Monitoring and Winterization – The Contractor shall monitor the operation of the system and carry out all minor repairs and required adjustments to the spray coverage of irrigation heads and operating times.

The irrigation system shall be properly winterized at the appropriate time of the season.

PAYMENT

766.91 General – Payment for the supply and installation of the irrigation system will be at the lump sum bid. The lump sum price bid shall be full compensation for all labour and equipment required for the specified preparation, trenching, installation, connection, testing, backfilling, clean-up, preparation of as-built drawings, and instruction in the proper use of the equipment and for all incidental work not required to be separately paid for.

APPENDIX 766-A

IRRIGATION SYSTEMS PRESSURE LOSS CALCULATION

DATA SHEET

This design is based on _____ GPM / L/min at _____ psi / kPa at the base of the sprinkler heads or downstream of the automatic valves in the case of drip irrigation systems.

The system is designed around _____ company product lines.

The static water pressure is _____ psi / kPa.

Dynamic water pressure is _____ psi / kPa.

Designed precipitation rate is _____ inches of water per hour / mm of water per hour.

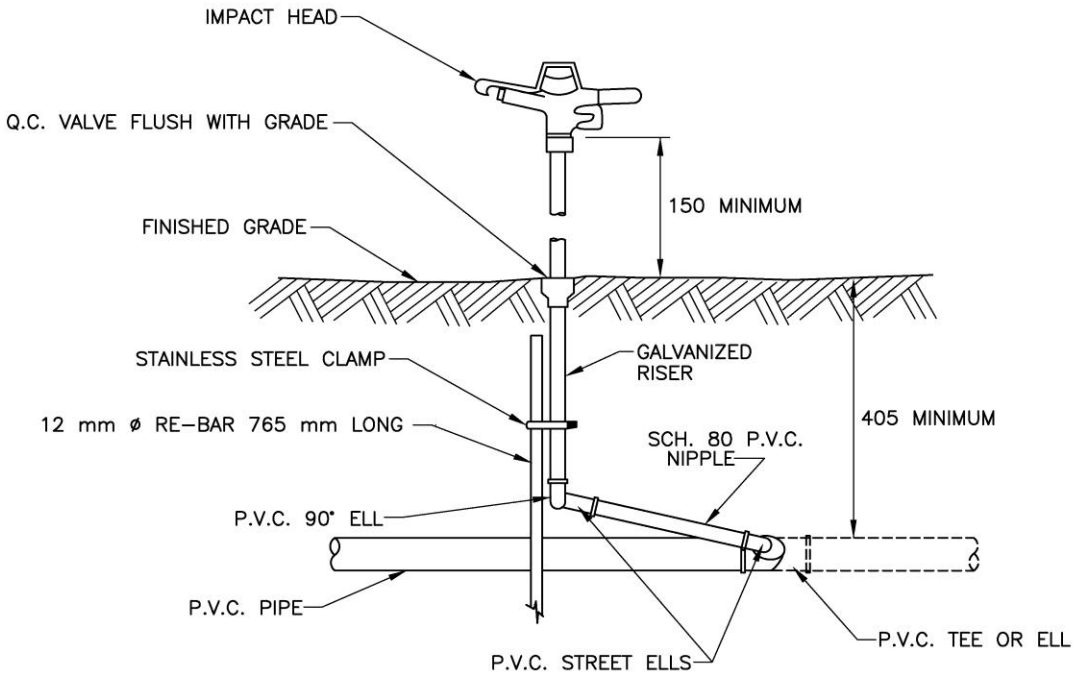
Designed operational time per zone is _____ minutes per day x _____ days per week.

Table 766-B: Irrigation System Pressure Loss

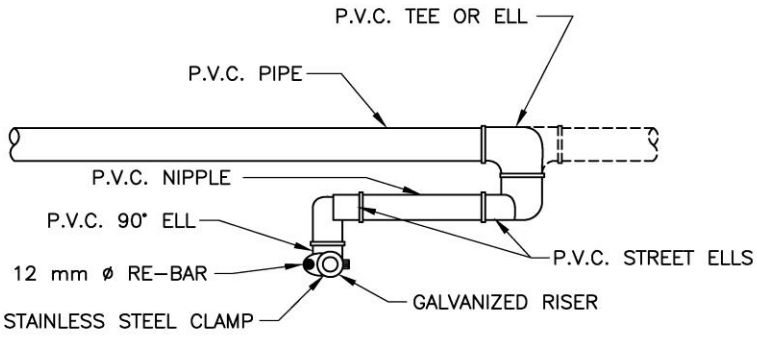
	Size	Length (ft / m)	Material Type	Manufacturer	Friction Loss/Gain (psi / kPa)
Service Line				—	
Water Meter		—	—	—	
Manual Valve		—			
Shutoff Cock		—			
Backflow Prevention Device		—	—		
Strainer		—	—		
Pipe/Fittings within the Backflow Prevention Chamber				—	
*** Worst Case Mainline (accumulated)**	—			—	
*** Worst Case Automatic Valve**		—			
*** Worst Case Lateral Length (accumulated)**				—	
Elevation Loss/Gain Throughout System	—		—	—	
Additional Valving	—	—			
				TOTAL FRICTION LOSS THROUGH SYSTEM	

* **Worst Case** is defined as the zone furthest away from the point of connection demonstrating the highest accumulated friction loss. **Accumulated friction loss** is defined as the aggregate of all friction loss found from the point of connection to the furthest sprinkler head in the zone or to the furthest emitter in a drip irrigation zone.

SWING JOINT DETAIL – IMPACT HEAD	SP766–01



ELEVATION



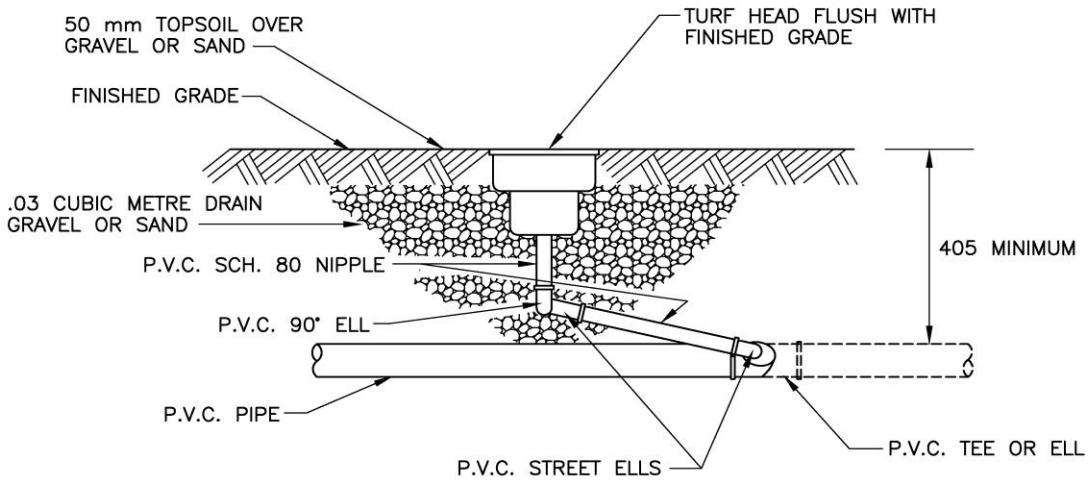
PLAN

SHOWN FOR IMPACT HEAD
QUICK–COUPLING VALVE SIMILAR

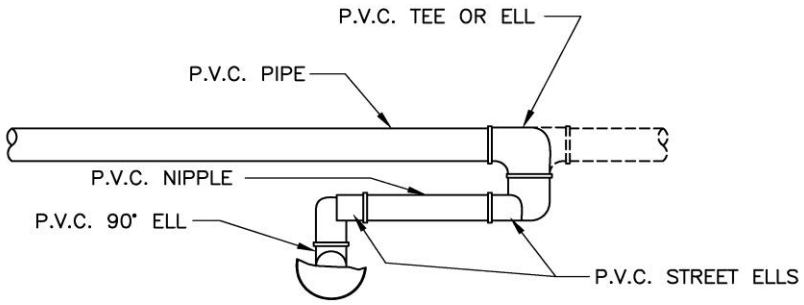
NOTE:
USE TEFLON TAPE ON ALL THREADED CONNECTIONS

NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

SWING JOINT DETAIL – TURF HEAD	SP766–02



ELEVATION



PLAN

SHOWN FOR TURF HEAD
LAWN HEAD SIMILAR

NOTE:
USE TEFLON TAPE ON ALL THREADED CONNECTIONS

NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

SECTION 769

PROTECTION AND RETENTION OF VEGETATION

DESCRIPTION

769.01 Scope – This Section refers to the protective measures required to safeguard vegetation from construction operations, equipment and vehicles, where vegetation is not designated for removal under the Contract, and covers the installation of barriers.

769.02 Related Work

(a) SS 165, Protection of the Environment;

(b) SS 200, Clearing and Grubbing;

(c) SS 201, Roadway and Drainage Excavation;

(d) SS 751, Topsoil and Landscape Grading.

769.03 Definitions

(a) **Barrier** – means fence consisting of approved material, supported by steel posts and being a minimum of 2.0 m high, without breaks or unsupported sections.

(b) **Dripline –** means the location on the ground surface directly beneath a theoretical line described by the tips of the outermost branches of trees.

(c) **Native Vegetation –** means areas of existing and/or indigenous shrubs, trees and groundcover.

(d) **Specimen Trees –** means trees so designated in the Contract Documents.

MATERIALS

769.11 Water – Water shall be free of impurities that would inhibit germination and growth or may be harmful to the environment.

The Contractor shall supply the water.

769.12 Fertilizer – Fertilizer shall be supplied to the specifications of the Special Provisions and to SS 751.18.

CONSTRUCTION

769.31 Operational Constraints

769.31.01 Vegetation designated for Retention – The Contractor's operations shall not damage vegetation designated for retention.

(a) Existing Native Vegetation shown on the Drawings or designated by the Ministry Representative to be retained shall be marked by the Contractor and inspected by the Ministry Representative, who may adjust the limits. A minimum of 48 hours notice shall be given to the Ministry Representative for this purpose.

(b) Areas shown on the Drawings adjacent to streams or as designated by the Ministry Representative as "Vegetation to Remain" areas are not to be disturbed, cleared or logged.

(c) The Contractor's operations shall not cause flooding, sediment deposits or deposition of debris in "Vegetation to Remain" areas.

(d) Where construction procedures substantially alter natural drainage patterns, interim drainage or irrigation shall be provided as necessary to compensate for construction interference.

(e) Construction procedures, stockpiling of materials or debris burning or disposal shall not be undertaken adjacent to Specimen Trees and/or Native Vegetation retained.

(f) Unless the Contract requires work within the dripline of Specimen Trees, equipment shall not be operated within that dripline.

(i) When the Contract requires work within the dripline of Specimen Trees, operation of equipment within that dripline area shall be kept to the minimum necessary to perform the work required, as required by the Ministry Representative.

(ii) Tree roots shall be protected from compaction by temporary placement of hog fuel or other lightweight insulation material, as required by the Ministry Representative.

(g) Equipment or vehicles shall not be parked, repaired or refuelled, construction materials shall not be stored, and earth materials shall not be stockpiled within the dripline area of any Specimen Trees.

769.31.02 Knotweed – If Giant, Bohemian, Japanese or Himalayan Knotweed species are found within the project footprint, they shall not be disturbed or removed without an invasive species management plan and pre-approval by the Ministry Representative. Vehicles, equipment, or materials shall not be parked or stored in areas where knotweed species are present.

769.32 Clearing and Grubbing – No clearing and grubbing shall be conducted for a radius of 3 m from the trunks which lie on the edge of clearing and grubbing zones, or as directed by the Ministry Representative.

(a) Trees to be removed shall be felled toward the centre of an area, away from selectively cleared or retained vegetation.

(b) Where Drawings or Special Provisions call for areas of "Close Cut No Grubbing", existing trees within the removal zones shall be carefully and cleanly cut as close to the ground as possible, with the understory vegetation retained, and the root zone protected from excessive disturbance.

769.33 Barriers for Existing Vegetation Protection – Barriers for existing vegetation protection shall be erected prior to commencement of construction operations, at locations specified in the Drawings, to provide a continuous barricade between vegetation and the area of work.

(a) The barriers shall be maintained erect and in good repair throughout the duration of construction operations, and shall be removed upon completion of the work, and disposed of off site of the project by the Contractor.

(b) The barrier shall be placed at the dripline of trees or forest edges and Native Vegetation unless there is inadequate space to provide a 1.5 m buffer zone between the barrier and the limit of grading.

(i) The barrier shall be placed within the dripline if necessary to provide a buffer zone of up to 1.5 m.

(ii) Under no circumstance shall the barrier be placed less than 0.75 m from the circumference of the trunk.

(iii) When the trunks of trees are less than 4.5 m apart, the trees shall be considered a group, and the barrier shall be placed to form a continuous barricade as specified in the Drawings.

(c) A barrier is not required where an existing fence will serve the same purpose. At such locations, the barrier shall terminate at the existing fence so that a continuous barricade is provided between the trees and the area of work.

769.34 Pruning and Repair of Specimen Trees – Specimen trees and trees safeguarded by barriers shall be repaired in accordance with the following.

(a) One third of the tree branches shall be selectively removed to reduce transpiration and compensate for dieback of roots in fill conditions and damage to the root system in cut conditions.

(b) Within five calendar days of damage, branches 25 mm or greater in diameter that are broken as a result of the Contractor's operations shall be cut back cleanly at the break, or to within 10 mm of their base, if a substantial portion of the branch is damaged.

Cuts shall be perpendicular to the growth of the branch to avoid leaving sharp points.

(c) Roots 25 mm or larger in diameter that are exposed by the Contractor's operations, shall be cut back cleanly to the soil surface within five calendar days of exposure.

(d) Bark that is damaged by the Contractor's operations shall be neatly trimmed back to uninjured bark, without causing further injury, within five calendar days of damage.

MAINTENANCE

769.71 Watering – The retained Specimen Trees and Native Vegetation shall be watered a minimum of three times during the summer or as stated in the Special Provisions or as directed by the Ministry Representative. The area immediately below the tree crown shall be soaked sufficiently to reach the feeder roots.

769.72 Fertilizing – Where specified or directed by the Ministry Representative, fertilizer shall be applied at a rate of 50 g/mm of caliper to existing Specimen Trees to be retained. The caliper measurement shall be taken 0.3 m above the grade. The fertilizer shall be applied once early in the growing season unless specified otherwise.

MEASUREMENT

769.81 General – Measurement for barriers will be made in metres according to the length of barrier installed to protect vegetation.

PAYMENT

769.91 General – Payment at the Unit Price bid for barriers shall be full compensation for work described and all work subsidiary and incidental thereto for which separate payment is not elsewhere provided.

Compensation for all costs other than installation of barriers associated with the work of protecting and maintaining Native Vegetation and Specimen Trees shall be deemed to be included in the Contract prices for the various Items of the contract.

SECTION 904

TIMBER (TREATED AND UNTREATED)

904.01 General

904.01.01 Lumber Grading

(a) Unless specifically stated otherwise elsewhere in the Contract, all timber shall be graded in accordance with the current Standard Grading Rules for Canadian Lumber of the National Lumber Grades Authority (including any Errata).

(b) Each piece of timber should be marked with a grade stamp from a lumber grading agency accredited by the Canadian Lumber Standards Accreditation Bureau (CLSAB).

(i) Timber supplied without a grade stamp, or where the grade stamp is obscured, shall have a certificate from a lumber grading agency accredited by the CLSAB that verifies the grade and species.

(ii) Copies of the grading certificate shall be kept by the Contractor and provided to the Ministry Representative upon request.

904.01.02 Timber Species

(a) Timber species, grade and preservative treatment shall be as specified in the Contract Documents.

(b) Coastal Douglas Fir [*Pseudotsuga menziesii var. menziesii*] shall be used when Douglas Fir products are specified to receive preservative treatment.

904.02 Framing – All the cutting, boring, framing, match marking, etc. required on all timber shall be done by competent framers in a thorough manner, in accordance with good work practice.

(a) It shall be done such that surfaces in contact shall bear evenly and fully; no shims or open joints are permitted.

(b) All measurements shall be accurate. Gains and daps shall have plane, smooth surfaces.

904.03 Ministry Quality Assurance and Inspection – The Ministry will carry out Quality Assurance and also may elect to arrange for its own inspection, before or after shipment to the Site.

(a) The Contractor shall provide the necessary facilities to enable the Ministry Representative to expeditiously examine as many pieces as are deemed necessary.

(b) The Ministry inspection shall not relieve the Contractor of responsibility to provide lumber with grade stamps or certificates of compliance or prevent its subsequent rejection at the construction site if found to fail any requirements of this specification.

(c) All material rejected (after maximum allowances are reached per applicable standards grading rules) shall be replaced at the Contractor's expense including shipping charges and removal of rejected material at the Site.

(d) The Contractor shall provide seven (7) days notice to the Ministry Representative regarding intent to ship a unit or product and the unit or product shall be made available for inspection prior to loading and shipping.

(i) No material shall be shipped prior to inspection or until a release for shipment has been issued by the Ministry Representative.

904.04 Handling of Untreated Timber – All materials shall be handled with reasonable care. Timber shall not be bashed, bruised, gouged, cracked, split or otherwise damaged.

904.04.01 Minor Damage – Minor damage, in the opinion of the Ministry Representative, shall be repaired at the Contractor's expense.

904.04.02 More Extensive Damage – Materials with more than minor damage will be rejected and replaced entirely at the Contractor's expense including all shipping and disposal charges.

904.05 Treated Timber – Treated timber shall be treated and handled in accordance with SS 908, Preservative Treatment - Wood Products.

After treatment, timber with checks exceeding the limiting sizes in the NLGA Standard Grading Rules for Canadian Lumber for the grade specified on the purchase order, Work Order, Special Provisions or Drawings shall be rejected.

SECTION 905

TIMBER – GLUED LAMINATED

GENERAL

905.01 General – The requirements of the current CSA O122 - Structural Glued-Laminated Timber shall apply to all glued laminated members. Wood species shall be as specified, except that if the members are specified to be Douglas Fir - Larch, and to be pressure treated, they may only be of Coast Region Douglas Fir.

Cutting, framing and boring of glulam members to receive preservative treatment shall be done before treatment insofar as possible.

905.02 Prefabrication Meeting – The Ministry will at its discretion, convene a prefabrication meeting with the Contractor to confirm the Ministry's requirements and to review issues such as, but not limited to, schedule, lines of communication, fabricator and sub-fabricator scope of work, location of all work, procedures on quality control, procedures for testing, lumber grading, preservative treatment and any other specific requirements as it relates to the Work.

905.03 Classification – Unless otherwise specified, the appearance grade shall be "Industrial" and the service grade shall be "Exterior". The stress grade shall be as shown on the Drawings, Contract, purchase order or work order.

905.04 Quality Control – The Contractor shall implement a quality control program to meet the Contract requirements. The quality control plan shall be made available to the Ministry Representative for review.

(a) Laminations shall be grade-marked and the marking shall be visible until the glue has been applied.

(b) The shear tests described in Appendices "A" and "B" of CSA O122 shall be carried out by the Contractor.

(c) Vacuum-pressure cycle tests described in CSA O122 shall be carried out by the Contractor.

(d) The Contractor shall provide test results from the shear and vacuum-pressure tests to the Ministry Representative.

905.05 Quality Assurance – The Ministry will implement a quality assurance program by auditing the Contractor's quality control program and by inspection and testing at its discretion.

(a) The Contractor shall notify the Ministry Representative at least 14 days before fabrication has commenced.

(i) The Contractor shall allow the Ministry Representative access to all parts of the Work, and shall supply such information and assistance as is required.

(ii) All timber and all phases of the work including pressure treatment, if applicable, may be inspected by the Ministry Representative.

(b) The Ministry Representative shall be given 48 hours notice of commencement of gluing and pressure treating.

(i) If the schedule is subsequently changed, the Contractor shall provide the 48 hours notice from the time that the Ministry is notified of this change.

(ii) If the product is not available or is not sufficiently complete for inspection as notified, at the sole discretion of the Ministry Representative, the Contractor shall be charged stand-by costs for the Ministry's quality assurance inspectors.

(c) The Contractor shall provide the necessary facilities to enable the Ministry Representative to expeditiously examine as many pieces as are deemed necessary.

(i) No material shall be shipped from the glue laminating plant prior to inspection or before a release for shipment has been issued by the Ministry Representative.

(ii) Inspection by the Ministry of glued laminated timber before shipment shall not prevent its subsequent rejection at the construction site if found to fail any requirements of the Contract.

(d) All material rejected shall be replaced.

905.06 Incising – All surfaces of members shall be incised prior to pressure treatment.

905.07 Preservative Treated Glued Laminated Members – Preservative treatment and handling of preservative treated wood shall be in accordance with CSA O122 and SS 908.

(a) Members shall be protected to avoid damage due to handling: .

(b) All handling of treated timber with pointed tools shall be confined to end grain and done in a manner to avoid damage to the original pressure-treated surface. Dogs, hooks, peavies or other equipment shall not be used on the side surfaces of treated timber

(c) The Contractor shall field treat any damage of treated timber by the methods outlined in the current edition

of CSA O80 Wood Preservation or the AWPA Standards.

(i) For bridge components, creosote and copper naphthenate shall be the only permitted field treatment preservatives.

(d) At the discretion of the Ministry Representative, timber which cannot be repaired by the above methods will be rejected.

905.08 Storage of Glued Laminated Members – Members lying in storage shall be supported evenly on a flat surface. When stored for a prolonged time, they shall be gapped to permit air circulation and covered to prevent UV damage.

905.09 Working Drawings – Working drawings shall consist of the following:

- Shop drawings,
- Transportation details, and
- Erection drawings

905.09.01 General

(a) **Documents to be Sealed –** Transportation details and erection drawings shall be prepared and sealed by a professional engineer registered with the Association of Professional Engineers and Geoscientists of the Province of British Columbia (APEGBC).

(b) **Contractor Design –** When the Contractor is responsible for the design of items that are detailed on the shop drawings, the shop drawings shall be prepared and sealed by a professional engineer registered with the Association of Professional Engineers and Geoscientists of the Province of British Columbia (APEGBC).

(c) **Lateral Stability –** The Contractor shall be responsible for the lateral stability of members and the structure at all stages of fabrication, transportation, erection, and subsequent construction, removal, and demolition, and shall, at the Contractors expense, design and provide bracing as necessary until completion of the Work.

905.09.02 Shop Drawings

(a) **Content –** Shop drawings shall show all information and details needed for the fabrication of the members including, but not limited to, such items as:

(i) member shapes and dimensions,

(ii) camber diagram,

(iii) complete geometric information that member dimensions and shapes are based on,

(iv) connection details,

(v) material and product standards,

(vi) mark numbers and general arrangement of member locations,

(vii) details of attachments,

(viii) fastener details,

(ix) mass of members, special tolerances,

(x) special handling instructions,

(xi) lifting details and lifting locations.

(b) **Available Copies –** A copy of the shop drawings shall be available at all times at the location where the components shown on the drawings are being fabricated.

(c) **Changes –** Changes to the glue laminated timber from what is shown on the reviewed shop drawings, or repairs made during fabrication and/or construction, shall be indicated by the Contractor on a marked-up set of shop drawings in digital format and submitted to the Ministry Representative at the completion of the Work.

905.09.03 Transportation Details – Transportation details shall include such items as:

(a) Description of hauling and handling equipment

(b) Weight of members

(c) Length and height of loads

(d) Location and method of member support

(e) Details for handling, storing, and loading of members.

905.09.04 Erection Drawings

(a) **Content –** Erection drawings shall show in detail the method of erection including, but not limited to, the following:

(i) Erection procedures

(ii) Procedures for off loading of members upon delivery

(iii) Details for temporary storage and support of members on site prior to erection

(iv) Equipment to be used

(v) Layout or general arrangement drawing showing the layout of the members, equipment positioning, and access roads

(vi) Crane make, model, and capacity charts, boom length(s), crane placement, and access for transporting of members to crane(s)

(vii) Radii and loads for crane lifts

(viii) Rigging details

(ix) Mass of members, rigging and special installation equipment

(x) Details for installation and removal of all falsework, temporary supports, temporary bearings, bracing, guys, dead-men, and lifting devices

(xi) Attachments to the bridge members and bridge structure for temporary support and special launching equipment

(xii) Detailed description of sequence of operations

(xiii) Details for special installation equipment such as a launching truss, launching nose, head frames, spreader beams and rollers

(xiv) Details for installation of members onto the permanent bearings

(xv) Traffic control plan for roadway and rail traffic

(xvi) Details for protection of existing utilities affected by the erection procedures

(xvii) Layout and details of fall protection and their sequence of installation.

(b) The Contractor shall be responsible for the lateral stability of members in accordance with SS 905.09.01(c).

(c) Details – The erection drawings shall be complete in detail for all anticipated phases and conditions during erection and during the temporary support of members.

(i) The Contractor shall submit calculations, upon request, to the Ministry Representative that demonstrate that specified factored demand/capacity ratios or allowable stresses are not exceeded in members, falsework, temporary bracing and temporary supports, and any Contractor-designed members that are intended to remain in the completed structure, and that member capacities and final geometry will be correct.

(ii) These calculations shall be sealed by the professional engineer who sealed the erection drawings.

(d) Falsework – Falsework, temporary supports and temporary bracing shall meet the requirements of CSA S269.1, "Falsework and Formwork" and shall also meet all the requirements for falsework given in Clauses 20.17 to 20.26 inclusive of the WCB Occupational Health and Safety Regulation.

(e) Changes – A professional engineer registered with APEGBC shall be responsible for any field designs and any changes made to the erection procedures.

Field designs and changes to the erection procedures must be documented and sealed by the responsible professional engineer and must be available at the Site prior to the affected erection Work being carried out.

(f) Falsework Inspection and Certification – Immediately before placement of loading on falsework, the Contractor must ensure that the falsework is inspected and a sealed engineering certificate is issued by a professional engineer registered with APEGBC which:

(i) Indicates the specific areas inspected and

(ii) Certifies that the falsework has been erected in accordance with the latest approved erection drawings and supplementary instructions.

905.09.05 Submittals – The Contractor shall submit to the Ministry Representative one set of all working drawings in digital format.

(a) Prior to submission to the Ministry Representative, working drawings shall be reviewed and approved by the Contractor.

(i) By this review and approval, the Contractor represents that it has determined and verified all field measurements, field construction criteria, materials, and similar data, and that it has checked and coordinated each working drawing with the requirements of the work and the contract documents.

(ii) The Contractor shall indicate its review and approval by including on each drawing the date and signature of a person designated by the Contractor as being responsible for the Work.

(b) Working drawings shall be submitted at least fourteen days prior to the fabrication of the Work and shall be accompanied by a transmittal listing each of the drawings submitted.

(i) At the time of submission, the Contractor must notify the Ministry Representative in writing of any deviations in the shop drawings from the requirements of the contract documents.

(ii) Any Work done or materials ordered prior to the review of the working drawings shall be at the Contractor's risk.

(iii) The Ministry will review the drawings for general compliance with the contract requirements.

(c) If modifications to the drawings are required, the Ministry Representative will return one set of drawings, marked up, to the Contractor.

(i) The Contractor shall re-submit one complete set of the revised drawings in digital format to the Ministry Representative.

(ii) Any drawing that has changed from the version originally submitted shall be identified as such on a transmittal accompanying the revised drawing set.

(d) If no exceptions are taken to the drawings, the Ministry Representative will return one set of reviewed drawings to the Contractor.

(e) Shop drawings will not be reviewed without the transportation details and erection drawings applicable to the members in question.

(f) Erection will not be allowed to proceed without the Ministry Representative's review of the method proposed.

(g) Review of working drawings shall not relieve the Contractor of responsibility for carrying out the work in accordance with the Contract documents.

(h) If so agreed to in advance by the Ministry Representative, working drawings may be submitted in paper format. The reference to shop drawing submittal copies shall be increased to four paper copies in this case.

(i) At least 14 days before fabrication is to commence, or as otherwise requested by the Ministry Representative, the fabricator shall submit a schedule of fabrication to the Ministry Representative in the form of a Gantt Chart.

(i) At the discretion of the Ministry, the schedule shall be updated on no less than a monthly basis.

(ii) The schedule shall be made available to the Ministry Representative for reference and planning of inspections and progress reporting.

(iii) At the request of the Ministry Representative, the Contractor shall report any interim variations to the schedule.

905.09.06 Working Drawing Format

(a) Format – Drawing files shall be submitted in PDF format to print out on 11" x 17" size pages. PDF sets shall be created by "distilling" CAD sheets rather than by scanning paper plan sets.

(i) Drawings shall be black images on a white background.

(ii) Working drawings shall be in the same system of units as the design drawings.

(iii) Lettering for notes and dimensions shall be at least 2.5 mm and 4 mm for headings. Drawings shall be legible when printed on 11" x 17" sheets.

(iv) The resolution of drawings shall be such that the finest detail must be legible at full scale on a monitor without zooming in (1-in. width on an 11" x 17" sheet is 1 in. on the monitor).

(v) All PDF sheets shall be assembled within a single file ensuring that all sheets are rotated to a "ready to read" orientation within the PDF file set.

Generally, 11" x 17" plan sheets should be in landscape and 8½" x 11" note sheets in portrait, so that the majority of text is vertical.

PDF sheets shall be ready to print out on appropriately sized paper sheets with no additional formatting required by the viewer.

(b) Non-compliant Submissions – Drawing sets that are not legible or that do not conform to submission requirements will be returned to the Contractor without being reviewed and the Contractor shall submit four paper copies of the drawing set as substitutes.

(c) Large or Complicated Pieces – Drawings of large or complicated pieces, where it is not practical to show all details on an 11" x 17" sheet, may be submitted on full size paper drawings, in which case the Contractor shall submit four copies of the drawing set.

(i) When revisions to full size paper drawings are required, the Contractor shall supply four complete revised sets to the Ministry Representative

(d) Transmittal – The Contractor shall transmit working drawings through attachments to e-mail.

Unless otherwise agreed to by the Ministry Representative:

(i) electronic attachments to an email must total no more than 20 MB and must be submitted unzipped.

(Note: BC Government servers reject all e-mails with "zipped" attachments without notice to either the sender or intended recipients.)

(ii) electronic attachments greater than 20 MB in size shall be:

(A) sent in two parts by separate emails, denoting "1 of 2" and "2 of 2" in the subject lines after other required subject-line information; or

(B) sent via a document and data management service such as SharePoint®.

(1) If this process is used, the limitation on drawing file size is waived.

(2) The Contractor shall be responsible for setting up the service and appropriate folders within the

document and data management software and for providing access to these folders to the fabricator, Ministry Representative and the design engineer.

(3) The Contractor and the Ministry Representative shall send e-mail notification to each other whenever they post drawings to the document and data management folders.

905.11 PAYMENT

905.91 Supply and Fabrication – Payment for supply and fabrication of glued laminated timber will be made at the lump sum price bid. Payment shall be for quality control, working drawings, the supply and fabrication of all necessary work. Payment shall also cover storage as necessary.

905.92 Shipping and Erection – Payment for shipping and erection of glued laminated timber will be made at the lump sum price bid. Payment shall be for quality control and the loading, shipping and unloading of glued laminated timber. Payment shall also cover falsework if necessary and the erection of glued laminated timber.

SECTION 906

ROUND TIMBER PILES

Note: Round timber piles may be used in the completed Work only when explicitly identified in the Contract as an acceptable product.

906.01 Scope – This Section covers the quality and manufacture of untreated and pressure-treated round timber piles.

Where convenient, the requirements contained in this Section use the wording of the current CSA O56 Round Wood Piles, in particular for term definition (see APPENDIX 906-A) and the majority of measurement and material requirements.

Preservative treatment of piles shall conform to the requirements of SS 908.

906.02 Species

906.02.01 Untreated Piles – This specification does not restrict the species used for untreated piles except as may be specified in the Contract Documents.

906.02.02 Pressure Treated Piles – Species of piles to be pressure treated shall be as specified and restricted to those for which pressure treating specifications are included in CSA O80 Wood Preservation.

906.03 Pile Size – The size of a pile shall be designated by length, minimum and maximum butt diameter, and minimum tip diameter. Sizes of piles normally available are shown in Table 906-A. Diameters in Table 906-A are minimum, except for the tolerance permitted in SS 906.03.01(f).

Table 906-A: Sizes of Timber Piles

Pile Length (m)	Pile Size Designation Minimum Diameter at Extreme Butt or Large End (mm)					
	230	250	280	300	330	360
	Minimum Diameter at Small End Tip (mm)					
Up to 6.1	150	180	200	230	250	250
6.1 to 10.4	150	150	180	200	230	250
10.7 to 13.4	–	–	150	180	200	230
13.7 to 18.0	–	–	–	180	180	200
18.3 to 21.0	–	–	–	150	180	200
21.3 to 27.1	–	–	–	–	150	180
27.4 to 32.0	–	–	–	–	130	150

906.03.01 Pile Diameter

(a) All measurements of diameter shall be made inside the bark.

(b) The diameter of treated piles shall be measured after treatment.

(c) Butt diameters shall be measured at the extreme butt. Maximum diameter at the butt shall not exceed 500 mm for any pile size.

(d) Tip diameters shall be measured at the extreme tip.

(e) The diameter of a pile at a given point along its length shall be determined either by measuring the circumference at that point and dividing the result by 3.14 or by taking the average of the maximum and minimum diameters measured at the point in question.

(f) A variation of -13 mm in the diameter at the tip or at the butt but not in both, shall be allowed in not more than 25% of the piles within a given substructure element provided there are at least four piles in that element.

(g) If allowances are specified, they shall supersede SS 906.03.01(f).

906.03.02 Pile Length

(a) Pile lengths shall be measured in metres.

(b) A variation of ±0.15 m from designated lengths will be permitted, unless allowable over and under lengths are specified.

906.04 General Material Requirements – Piles shall be cut from sound live trees. Sides and end surfaces of the piles shall be free of ice, snow and mud, and exposed for visual inspection.

906.04.01 Untreated Piles – Untreated piles shall be sound and close-grained.

906.04.02 Treated Piles – Treated piles shall meet the following:

(a) have not less than 20 mm sapwood, and shall be cleanly peeled;

(b) in any 300 mm length of the pile, the maximum combined total depth of all trimmed scars shall not exceed 25 mm;

(c) not be cored until 24 hours after treatment; and

(d) be cored in the middle third of the length of the pile.

906.05 Defects

906.05.01 Prohibited Defects

(a) Through checks

(b) Bird holes

(c) Cross-breaks of grain (cracks)

(d) Decay

(e) Nails, spikes and other metal or foreign substance

(f) Holes in treated piles (except holes for test purposes, which shall be properly plugged). Holes for brailing purposes shall be drilled prior to treatment

(g) Shakes in the tip

(h) Splits in the tip

(i) Insect damage

(j) Any abnormal change in cross-section (including ground swell)

(k) Knot clusters

(l) Unsound scars (cat faces)

(m) Short crooks (see SS Drawing SP906-01)

(n) Reverse sweep (see SS Drawing SP906-02)

(o) Burst – unsound piling which have 15 mm or more of a concentration of oil in the deadwood

906.05.02 Permitted Defects

(a) Firm red heart

(b) Hard stain

(c) Compression wood

906.05.03 Limited Defects

(a) **Checks –** Checks caused by treating and normal seasoning shall not exceed the following:

(i) Checks in the tips of the pile from the pith to the circumference shall not be wider than 10 mm at the circumference.

(ii) If checks are wider than 10 mm but not deeper than 1/3 of the diameter of the pile, not more than three of these checks shall occur in the circumference of the pile at that point. If checks are deeper than 1/3 of the diameter of the pile but not to the heart, and not wider than 10 mm, then only two such checks shall occur in the circumference of the pile at that point. The sum of all the widths of checks in a pile shall not exceed 1/4 of the diameter where they occur.

(b) **Sound Knots –** Sound Knots will be permitted as follows:

(i) For piles 15 m and less in length, knots up to 100 mm in diameter will be permitted provided that the sum of all knot diameters does not exceed 1/3 of the diameter of the pile at the section where they occur;

(ii) For piles more than 15 m in length, knots between:

(A) mid-length and butt shall conform to the requirements for piles of 15 m or less;

(B) mid-length and the tip, up to 125 mm in diameter, will be permitted provided that the sum of all knots does not exceed 1/2 of the diameter of the pile at the cross-section where they occur.

(c) **Shakes –** Shakes in the butt end of a pile are permitted provided that the length of any single shake, or the total length of any number of shakes, measured along the line of the shake, does not exceed 1/3 of the measured diameter of the butt.

(d) **Splits –** Splits in the butt end of a pile are permitted provided that they are not longer than the diameter of the butt.

(e) **Sweep –** Sweep is permitted provided that it cannot be classified as short crook or reverse sweep and provided that:

(i) For piles less than 21 m in length, a straight line joining the mid-point of the butt and the mid-point of the tip does not at any intermediate point pass outside the surface of the pile;

(ii) for piles 21 m and up to 24 m in length, a similar straight line does not lie more than 25 mm outside the surface of the pile; or

(iii) for piles over 24 m in length, a similar line does not lie more than 50 mm outside the surface of the pile.

Note: See SS Drawing SP906-01.

(f) **Holes –** Holes for untreated piles less than 13 mm in average diameter will be permitted provided that:

(i) they are scattered over the surface of the pile; and

(ii) the sum of the average diameter of all holes in any square 300 mm x 300 mm of pile surface does not exceed 38 mm and the depth of any one hole does not exceed 38 mm.

(g) **Spiral Grain –** Spiral Grain shall not exceed 1/2 turn in any 6 m length of pile.

906.06 Manufacturing Requirements – All piles shall be cut above the ground swell and have a uniform taper throughout their whole length.

Completely overgrown knots rising more than 30 mm above the pile surface, branch stubs, and partially overgrown knots shall be trimmed close.

Peeling of piles by hand or machine shall be done carefully so as not to gouge or otherwise damage the surface of the pile, and the reduction in thickness of sapwood shall be the minimum possible.

Sawing of butts and tips shall be cut perpendicular to the axis of the pile. Bevelling at the ends of the piles shall not be permitted.

906.07 Storage – Piles shall be stacked in treated or other non-decaying skids of such dimensions, and so arranged as to support the piles without producing noticeable distortion of any of them. A cover should be put over them to protect against the elements.

906.08 Handling – Handling of preservative treated piles shall be in accordance with SS 908.

906.08.01 Additional Handling Requirements – Handling shall also meet the following:

(a) All piles shall be handled with reasonable care. Piles shall not be bruised or otherwise damaged. Minor damage shall be repaired and severely damaged piles shall be replaced. Handling damage will be unacceptable if it reduces the depth of sapwood so as to render the pile untreatable.

(b) During lifting, long piles shall be supported at a sufficient number of properly located points to prevent damage due to excessive bending.

(c) Dogs, hooks, peavies or other equipment shall not be used on the round surface of treated piling, except in the last one metre of the butt end.

(d) The Contractor shall make good any superficial damage of treated piles by field treatment methods in accordance with CSA O80 Wood Preservation. Piles, which, in the opinion of the Ministry Representative, cannot be made good by such methods, will be rejected.

906.09 Quality Control – Quality control inspection of any preservative treatment shall be in accordance with SS 908.

Treated piles being supplied from previously treated stock shall have been clearly stamped with the applicable Charge Number or other positive identification, and the appropriate treatment records shall be made available to the Ministry Representative for inspection.

All piles being supplied from previously treated stock shall be subject to re-inspection which shall be in accordance with SS 908.

Piles to be supplied from previously treated stock which have been stored in water shall, if ordered by the Ministry Representative, be removed from the water if such action is required to provide adequate inspection. At the Contractor's option, such inspection may be carried out at the Site. All piles rejected at the Site shall be removed and replaced.

906.10 Quality Assurance – The Ministry will implement a quality assurance program by auditing the Contractor's quality control program and by inspection and testing at its discretion.

Piles may have quality assurance inspection before shipment to the Site, at the Site or both. The Contractor shall provide the necessary facilities to enable the Ministry Representative to expeditiously examine all parts of each pile. All piles rejected shall be replaced at the Contractor's expense, including shipping charges and removal of reject piles from the Site, if applicable. Inspection of the piles before shipment shall not prevent their subsequent rejection at the Site if found to fail any requirements of this Section.

APPENDIX 906-A

DEFINITIONS

Check: Separation of the wood along the grain, the greater part of which occurs across the rings of annual growth.

Through check: A check, which extends from surface to surface of the pile, usually through the pith centre.

Compression wood: Wood which has grown abnormally as often occurs on the lower side of branches and inclined trunks of coniferous trees.

Compression wood:

- is denser and harder than normal wood but may be lower in strength for its mass.
- is characterized by relatively wide annual rings, usually eccentric.
- has a relatively high proportion of summerwood (frequently more than 50% of the width of the annual rings in which it occurs).
- exhibits little contrast in colour between springwood and summerwood, and
- shrinks excessively lengthwise as compared with normal wood.

Crack: Separation of the wood cells across the grain (this may be due to internal strains resulting from unequal longitudinal shrinkage or to external forces).

Decay (rot, dote): The disintegration of the wood substance, due to the action of wood-destroying fungi (rot and dote mean the same as decay).

Grain: The direction, size, arrangement, and appearance of fibres in a pile.

Spiral grain: A type of growth in which the fibres take a spiral course about the bole of a tree instead of the normal vertical course. The spiral may extend right-handed or left-handed around the tree trunk. The amount of spiral grain in a pile is measured as the distance in metres, along the axis of the pile in which one complete twist of the spiral occurs, and is expressed as, for example, one (1) turn in 12 m.

Hole: An opening, perforation or cavity in a pile.

Bird holes: holes and damage caused by woodpeckers and other species of birds.

Plugged holes: holes that have been filled by forcing in another piece of wood.

Insect damage: Result of boring in the pile by insects or insect larvae. Scoring or channelling of the surface is not classified as insect damage.

Knot: That portion of a branch or limb that has become incorporated into the body of a tree.

Knot cluster: Two or more knots grouped together as a unit with the fibres of wood deflected around the entire unit.

Knot diameter: The diameter of a knot as it appears on the surface of a pile measured in a direction at right angles to the lengthwise axis of the pile.

Unpeeled: No attempt is made to remove the bark from the pile, but does not mean that all of the bark is on the pile as it may be scuffed, knocked, or worn off after the pile is cut.

Rough peeled: All of the outer bark is removed from the pile.

Clean peeled: All of the rough bark is removed from the pile, and all of the inner bark from at least 80% of the surface of the pile, no piece of inner bark more than 200 mm long remains on the surface of the pile, and there is at least 30 mm of clean wood surface between any two strips of inner bark.

Red heart: A condition caused by fungus (*Fomes pini*, *Trametes pini*) which occurs in the living trees of some conifers. It is characterized in the early stages of infection by a reddish or brownish colour in the heartwood. Later the wood in the living tree disintegrates (decays) in small, usually distinct areas that develop into white-lined pockets.

Firm red heart: the early stages of infection, characterized by a reddish or brownish colour in the heartwood, which does not affect the strength of the pile.

Scar (cat face): A depression in the surface of the pile resulting from a wound where healing has not re-established the normal cross-section of the pile.

Shake: A separation along the grain, the greater part of which occurs between the rings of annual growth.

Short crook: A localized deviation from straightness which, within any section 1.5 m or less in length, is more than 65 mm (see SS Drawing SP906-01).

Sound: Free from decay and insect holes.

Split: A lengthwise separation of the wood due to the tearing apart of the wood cells which usually extends from surface to surface of the pile.

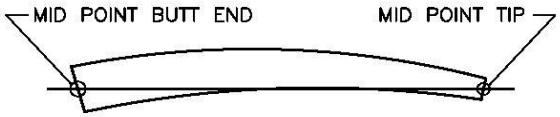
Stain: A discolouration on or in the wood other than its natural colour, caused by the action of certain moulds and fungi.

Hard stain: A stain, which is not accompanied by softening or other disintegration of the wood.

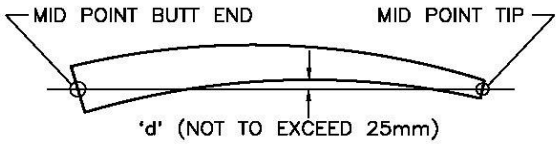
Sweep: The deviation of the pile from straightness (see SS Drawing SP906-01).

Reverse sweep: A deviation from straightness, which changes direction or reverses direction in the length of the pile (see SS Drawing SP906-02).

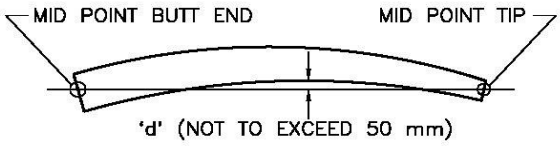
MEASUREMENT OF SWEEP AND CROOK IN PILES — SP906–01



LESS THAN 21 m

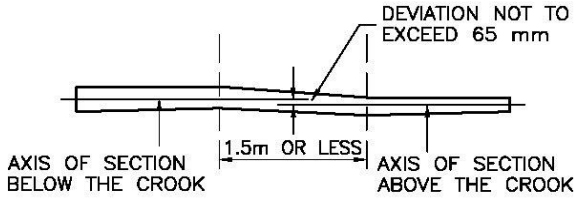


21 TO 24 m

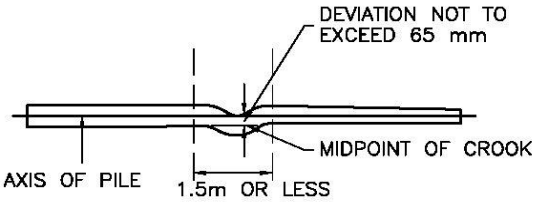


OVER 24 m

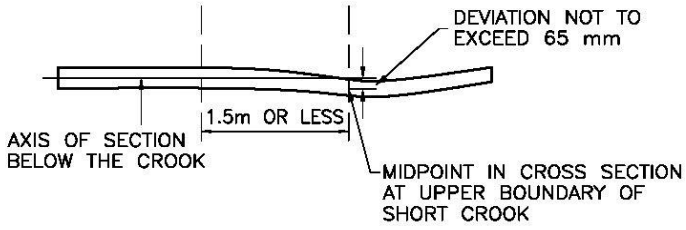
MEASUREMENT OF SWEEP



CASE 1: WHERE THE REFERENCE AXES ARE APPROXIMATELY PARALLEL.



CASE 2: WHERE AXES OF SECTIONS ABOVE AND BELOW THE CROOK COINCIDE OR ARE PRACTICALLY COINCIDENT.



CASE 3: WHERE AXIS OF SECTION ABOVE SHORT CROOK IS NOT PARALLEL OR COINCIDENT WITH AXIS BELOW THE CROOK.

MEASUREMENT OF SHORT CROOK

NOTE:
THE THREE CASES SHOWN ARE TYPICAL AND INTENDED TO ESTABLISH THE PRINCIPLE OF MEASURING SHORT CROOKS. THERE MAY BE OTHER CASES NOT EXACTLY LIKE THOSE ILLUSTRATED.

NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

REVERSE SWEEP IN PILES	SP906–02



REVERSE SWEEP IN PILES

NOTES: WHEN CONDITIONS OF DESIGN REQUIRE, THE FOLLOWING REQUIREMENTS APPLY:

1. FOR PILES LESS THAN 10 m IN LENGTH, A STRAIGHT LINE CONNECTING THE MID–POINT AT THE BUTT AND THE MID–POINT AT THE TIP SHALL NOT LIE LESS THAN 75 mm INSIDE THE SURFACE OF THE PILE;
2. FOR PILES 10 m AND UP TO 12 m IN LENGTH, A SIMILAR STRAIGHT LINE SHALL NOT LIE LESS THAN 50 mm INSIDE THE SURFACE OF THE PILE;
3. FOR PILES 12 m AND UP TO 15 m IN LENGTH, A SIMILAR STRAIGHT LINE SHALL NOT LIE LESS THAN 25 mm INSIDE THE SURFACE OF THE PILE; OR
4. FOR PILES OVER 15 m IN LENGTH, A SIMILAR STRAIGHT LINE SHALL NOT PASS THROUGH THE SURFACE OF THE PILE.

NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

SECTION 908

PRESERVATIVE TREATMENT – WOOD PRODUCTS

908.01 Scope – This Section covers the quality and requirements for preservative treatment of wood products using pressure treatment.

908.02 General – Treated wood shall meet the requirements of this SS 908 as well as the product specific requirements in, without limitation, the following related Standard Specification sections, as applicable to the intended use:

(a) SS 213 Timber Bridges – Construction

(b) SS 214 Timber Piling Construction

(c) SS 604 Steel Traffic Barrier Construction

(d) SS 635 Electrical and Signing

(e) SS 700 Wildlife Exclusion Fencing

(f) SS 741 Fence Construction

(g) SS 904 Timber (Treated and Untreated)

(h) SS 905 Timber – Glued Laminated;

(i) SS 906 Round Timber Piles

(j) SS 909 Treated Wood Fence Posts

908.03 References – The standards, specifications and publications as listed in Table 908-A shall apply to all treated wood products.

When a requirement of the CSA O80 Series of Standards conflicts with a requirement of an AWPA Standard, the requirement of the CSA O80 Series of Standards shall take precedence.

908.04 Timber Grading – All timber (excluding round fence posts) shall be graded in accordance with the National Lumber Grades Authority's current Standard Grading Rules for Canadian Lumber.

(a) Each piece of timber (excluding fence posts) should be marked with a grade stamp from a lumber grading agency accredited by the Canadian Lumber Standards Accreditation Bureau (CLSAB).

(b) Timber supplied without a grade stamp, or where the grade stamp is obscured, shall have a certificate from a lumber grading agency accredited by the CLSAB that verifies the grade and species.

(i) Copies of the grading certificate shall be kept by the Contractor and provided to the Ministry Representative upon request.

908.05 Species and Grade – Timber species and Grade shall be as specified in the related Standard Specification sections and Contract Documents.

"Coastal" Douglas Fir [*Pseudotsuga menziesii var. menziesii*] shall be used when Douglas Fir products are specified to receive preservative treatment.

Table 908-A: Standards, Specifications, and Publications

Agency / Spec.	Description
CSA	**Canadian Standards Association**
O80	Wood Preservation Series of standards
O80.0	General Requirements for Wood Preservation
O80.1	Specification of Treated Wood
O80.2	Processing and Treatment
O80.3	Preservative Formulations
AWPA	**American Wood Protection Association**
M1	Standard for the Purchase of Treated Wood Products
M2	Standard for the Inspection of Preservative Treated Products for Industrial Use
M3	Standard for the Quality Control of Preservative Treated Products for Industrial Use
T1	Use Category System: Processing and Treatment Standard
U1	Use Category System: User Specification for Treated Wood
ALSC	**American Lumber Standard Committee** Treated Wood Program
CLSAB	**Canadian Lumber Standards Accreditation Board**
CWPCA	**Wood Preservation Canada**
NLGA	**National Lumber Grades Authority** Standard Grading Rules for Canadian Lumber
WWPI	**Western Wood Preservers Institute** Best Management Practices for Use of Wood in Aquatic and Other Sensitive Environments

908.06 Conditioning – All timber shall be air seasoned sufficiently and properly conditioned in accordance with CSA O80 prior to treatment to ensure proper penetration of preservative during the treatment process and to prevent checking after treatment.

In exceptional circumstances, the Ministry Representative may permit conditioning by steaming as per CSA O80.2, clause 4.4.4 and/or the latest AWPA T1 Standard.

908.07 Preservative Treatment – All treated wood materials shall be pressure treated in accordance with CSA O80.

The preservative treatment of laminated veneer lumber shall be in accordance with AWPA U1 and parallel strand lumber shall be in accordance with AWPA U1 and AWPA T1.

908.07.01 Acceptable Preservatives – One of the following preservatives shall be used:

(a) creosote;

(b) pentachlorophenol in Type A hydrocarbon solvent;

(c) copper naphthenate in Type A hydrocarbon solvent;

(d) chromated copper arsenate, Type C (CCA);

(e) ammoniacal copper zinc arsenate (ACZA);

(f) alkaline copper quaternary, Type A (ACQ-A);

(g) alkaline copper quaternary, Type C (ACQ-C);

(h) alkaline copper quaternary, Type D (ACQ-D); or

(i) copper azole Type B (CA-B).

908.07.02 Certified Treatment Facilities – All pressure treated material shall come from a treatment plant/facility that is certified under and in compliance with the Canadian Wood Preservation Certification Authority (CWPCA) program.

A copy of the CWPCA certificate and certification letter must be submitted to the Ministry Representative upon request.

908.07.03 Environmental Compliance – Preservatives shall comply with all required environmental regulations. Treated wood for use in bridges or for use near or in aquatic environments shall be treated in accordance with the most recent version of Best Management Practices for Use of Wood in Aquatic and Other Sensitive Environments, published by Western Wood Preservers Institute et al (WWPI BMPs).

908.07.04 Other Conditions – The type of preservative, conditioning, treatment, penetration and retention for the treated wood product shall be appropriate for the species, size, and end use of the product.

All sawn wood and glued-laminated members shall be incised before treatment in accordance with CSA O80.

All treated wood shall be substantially devoid of free surface preservative liquid and preservative deposits.

908.07.05 Use Category – The Use Category and the type of preservative for treated wood materials shall be in accordance with the following requirements. If there is any conflict between any of these requirements, the following shall apply in the descending order of precedence:

(a) Special Provisions;

(b) Drawings;

(c) Related Standard Specifications as listed in SS 908.02;

(d) This SS 908 and Table 908-B; and lastly

(e) CSA O80, Table 2 "Use categories for specific products, uses and exposures".

908.07.06 Pre-cut or Field Treatment

(a) Cutting, framing, and boring of timbers to receive preservative treatment shall be done before treatment insofar as possible.

(b) In the event that cutting or drilling becomes necessary after treatment, a field treatment preservative specified in CSA O80 or the AWPA M4 shall be used and applied in accordance to its label.

  (i) For bridge components, creosote and copper naphthenate shall be the only permitted field treatment preservatives.

(c) At least two coats shall be applied and where possible, the colour of the preservative treatment used for protecting field cuts shall match the original preservative treatment colour.

908.08 Hardware, Fasteners and Metalwork

908.08.01 General – All hardware, fasteners and metal work in contact with treated wood products used in permanent structures shall be stainless steel or hot-dipped galvanized in accordance with the ASTM A123, ASTM A153 (Class D), or ASTM A653 (G90 coating class) as applicable.

908.08.02 Wood Treated with CCA, ACZA, ACQ-A, ACQ-C, ACQ-D, or CA-B – All hardware, fasteners and metal work used in permanent structures shall be hot-dipped galvanized in accordance with the ASTM A123, ASTM A153 (Class D), or ASTM A653 (G185 coating class) as applicable. Nails, spikes and sheet metal fastenings shall be 304 or 316 stainless steel when specified.

908.08.03 Galvanized fastenings – Galvanized nuts shall be retapped to allow for the increased diameter of the bolt due to galvanizing. Heat-treated alloy components and fastenings that may be affected by the heat of the zinc bath shall have corrosion protection provided by an alternate means as approved by the Designer and the Ministry Representative.

Table 908-B: Use Category

Type of Wood Product	Use Category per CSA O80
All treated wood products exposed to brackish water or salt water	UC 5A
Cribbing, ballast walls	UC 4.2
Timber for bridge barrier and railings	UC 4.2
Timber for bridge decks, floor beams, stringers *	UC 4.2
Glued-Laminated Timber (see SS 905)	UC 4.2
Round Timber Piles, Pile Cap Beams and Pile Bracing (see SS 906)	UC 4.2
Sign Posts and associated components (see SS 635)	UC 4.1
Roadside guardrail posts and other guardrail components (see SS 312 and SS 604)	UC 4.2
Pedestrian and cyclist railings (excluding bridge barrier and roadside guardrail)	UC 4.1 (using preservative safe for pedestrian contact)
Wood Fence Posts and associated wood fence components (see SS 700, SS 741, and SS 909)	UC 4.1

* Bridge deck running planks do not require preservative treatment

908.09 Handling of Treated Wood Products – All treated wood materials shall be handled with reasonable care to prevent damage of the pressure-treated surface such as puncture, cutting or crushing of fibre.

(a) Unloading – When unloading at the Site, the various pieces shall be sorted and stacked in such a manner as to require a minimum of re-handling before being placed in the structure.

(i) The Contractor shall make good, at the Contractor's expense, any damage to treated wood materials supplied and/or supplied by the Ministry after having taken delivery of same.

(b) Minor Damage – Subject to the approval of the Ministry Representative, the Contractor shall treat any small areas of superficial damage of treated wood materials using the field treatment methods specified in CSA O80 or AWPA M4.

(i) For bridge components, creosote and copper naphthenate shall be the only permitted field treatment preservatives.

(ii) At least two coats shall be applied and where possible, the colour of the preservative treatment used for repair shall match the original preservative treatment colour.

(c) More Extensive Damage – Severely damaged pieces and those which cannot be made acceptable by the above methods will be rejected at the sole discretion of the Ministry Representative.

908.10 Quality Control at Treatment Plant

908.10.01 General – A quality control program shall be implemented at the treatment plant in accordance with CSA O80, CWPCA, and AWPA M3. Treated products shall be inspected in accordance with AWPA M2.

The quality control program shall include conducting and applying all quality control steps and inspections, including third-party inspections, as per industry standards and as required to meet these specifications. The quality control program shall maintain and properly document all records of quality control inspection reports, worksheets, checklists, testing and data verification and types of treatments.

Quality control shall include plant inspection of treated wood products before, during, and after preservative pressure treatment to ensure the final treated wood products meet these specifications.

Copies of any quality control records for any treatment charge and treatment test shall be made available to the Ministry Representative upon request.

908.10.02 Third Party Inspection – Quality control shall include third-party inspection. Third-party quality control inspections must be performed at the treatment plant.

(a) The third-party inspection agency shall meet the one of the following requirements:

(i) Accreditation by the American Lumber Standard Committee (ALSC) under their treated wood program, or,

(ii) Experience acceptable to the Ministry Representative and the Contractor.

(b) The third-party inspection agency and the Ministry Representative shall have safe access to all parts of the treatment facility and shall be supplied information and assistance as required.

(c) The third-party inspection agency shall provide an inspection report that confirms the treated wood conforms to the preservation requirements of CSA O80 and/or where applicable AWPA U1 and AWPA T1 and/or the WWPI BMPs.

Inspected material meeting these requirements shall be stamped with a certification mark.

(d) Alternatively, for round fence posts or when otherwise acceptable to the Ministry Representative, the third-party inspection agency shall provide a certificate of compliance that confirms that the inspected material has been treated in accordance with CSA O80.

(e) Inspection reports must indicate both the "Use Category" and the type of preservative used.

(f) Inspection reports shall be submitted to the Ministry Representative prior to the material being incorporated into the Work and shall be made available to the Contractor and the treatment facility.

(g) No material shall be shipped prior to completion of the third-party inspection indicating that the treated products meet the specified requirements.

908.11 Quality Assurance – The Ministry may implement a quality assurance program by auditing the Contractor's quality control program and by inspection and testing at its discretion.

(a) The Ministry reserves the right to conduct whatever tests are deemed necessary to confirm that the material complies with the requirements of this specification before a release for shipment is issued.

(i) The Ministry may elect to arrange for its own quality assurance inspection of any products by a lumber grading agency accredited by the CLSAB, the ALSC, and/or a third party inspection company, before or after shipment to the Site.

(ii) QA may be performed on the wood product or treatment products at any time, including before, during, and after treatment.

(b) The Contractor shall provide the necessary facilities to enable the Ministry Representative to expeditiously examine as many pieces as are deemed necessary.

(c) The Ministry Representative or delegate shall be granted complete access to the treatment plant and all its facilities, at any time during business hours, to conduct quality assurance inspections and testing.

(d) The Contractor shall notify the Ministry Representative at least ten (10) full working days before the material is ready for quality assurance inspection prior to shipment.

(e) The Ministry inspection shall not relieve the Contractor of responsibility to provide fully compliant treated wood products, including grade stamps or certificates of compliance.

908.12 Approval of Materials – Inspection of treated wood products before shipment shall not prevent its subsequent rejection at the Site if found to fail any requirements of the specifications.

Rejected material (after maximum allowance is reached per CSA O80) shall be replaced at the Contractor's expense including shipping charges and removal of rejected materials if applicable.

Any rejected material shall be removed from the Site and replaced at the Contractor's expense immediately after notification.

Ministry costs for re-inspection or re-testing for rejected material shall be at the Contractor's expense.